



Fig. 6: A bright image

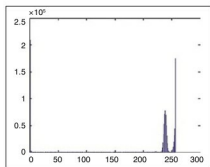


Fig. 7: Histogram of bright image



Fig. 8: Dark image

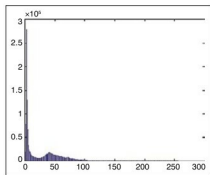


Fig. 9: Histogram of dark image

in Fig. 7. Bright image means all its pixels have a value close to 255. As can be seen from Fig. 7, most of the histogram is towards the right side as

discussed earlier.

Dark image. Next, we take the example of a very dark image as shown in Fig. 8.

It can be observed from Fig. 8 that dark image corresponds to the pixel values close to zero, hence it can be seen from its histogram (refer Fig. 9) that most of the pixels are close to zero, resulting in a histogram towards the left side.

Uniform image. For the third example, a uniform bright image is shown in Fig. 10. The image is well captured with

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com



Fig. 10: Uniformly bright image

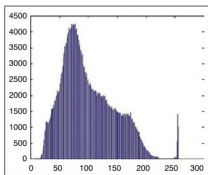


Fig. 11: Histogram of uniformly bright image

uniform lighting condition.

By looking at the histogram in Fig. 11 it can be seen that for a good image it is nearly uniformly distributed. This

means all the gray values have nearly equal contribution to create a good and clear image.

So, if you want to capture an image of high quality, just look at the histogram of the image generated by the camera and try to adjust the lighting or position of the camera in such a way that the histogram is uniformly distributed.

Software program

MATLAB program is used here to create histogram of an image. Though we have an in-built function in MATLAB to do this job, but the whole idea is to understand the concept of histogram in detail. It is only then we can optimise the code to make it work faster for the coming generation of high-end cameras. The MATLAB code is as listed below:

```
y=imread('Path of image to be inserted');
[m,n,num_colors]=size(y);
if num_colors> 1
    x1=rgb2gray(y);
    x=double(x1);
else
    x=double(y);
end
counter=zeros(256,1);
for i=1:size(x,1)
    for j=1:size(x,2)
        for k=0:255
            if (x(i,j)==k)
                counter(k+1)=counter(k+1)+1;
            end
        end
    end
end
bar(counter,0.00001),title
('HISTOGRAM');
```

Before compiling the code, you need to provide the path of the image in the code. When you compile and run the code in MATLAB, you will get the histogram of the image file. ●

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